

AIGR Focused Hypothesis Report —

AIM33 (Q04516) / GO:0004128

Gene: AIM33 (YML087C), *Saccharomyces cerevisiae* (NCBITaxon:559292) **UniProt:** Q04516 — "Uncharacterized oxidoreductase AIM33" **Focus:** function_assignment · slug `function-hypothesis-go-0004128` **Seed hypothesis:** AIM33 has cytochrome-b5 reductase activity, acting on NAD(P)H (GO:0004128) **Annotation under review:** GO:0004128, evidence IBA, ref GO_REF:0000033

Executive Judgment

Verdict: PARTIALLY SUPPORTED (plausible family-level inference; unconfirmed for AIM33 specifically; not paralog over-annotation).

The molecular-function *class* is well supported: AIM33 has the complete domain architecture of the flavoprotein pyridine-nucleotide cytochrome reductase family (FAD-binding FR-type + NAD_binding_1), retains both catalytic cofactor fingerprints (NAD-binding `CGPPKM`; FAD/NAD pyrophosphate `GSGITP`), and folds confidently into an FNR-like reductase module (AlphaFold mean pLDDT = 87.25). Critically, the closest experimentally-characterized member of AIM33's own PANTHER subfamily (SF143), **PGA3/Q12746**, is a bona fide **NADH-dependent cytochrome b5 reductase** (it is also the IBA seed). So the transfer of GO:0004128 to AIM33 is a *defensible, standard phylogenetic inference*, **not** a case of paralog over-annotation onto an unrelated protein.

However, the specific term is **unconfirmed for AIM33**: SGD explicitly curates its molecular function as **ND ("No biological Data")**, there is no direct enzymatic assay, AIM33 sits in a distinct subfamily whose physiological acceptor may be **coenzyme Q6** rather than cytochrome b5, and its topology (3 TM helices) differs from classic soluble/mono-anchored cyt-b5 reductases. The co-annotated **cellular-component (plasma membrane) and process (ergosterol biosynthesis)** IBA terms are PGA3-specific carry-overs that **conflict** with AIM33's only experimental phenotype (increased mitochondrial genome loss).

Bottom line for the curator: Retaining GO:0004128 as an **IBA** MF term is methodologically acceptable, but it should be treated as *inferred/unconfirmed* (consider the generalized parent MF as the safest evidence-backed statement). The higher-priority curation action is on the **plasma-membrane CC term**, which is likely mis-transferred for AIM33.

Evidence Matrix

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Co lim
UniProt Q04516 / QuickGO	review/ database	qualifies	Is GO:0004128 experimentally supported for AIM33?	Only IBA (GO_REF:0000033) + automated IEA; SGD MF = ND ; IBA seeds = MCR1 + PGA3	<i>S. cerevisiae</i> , curated DB	Hi inf no exi
InterPro/Pfam (PF00970 FAD_binding_6; PF00175 NAD_binding_1; IPR001709)	structural/ evolutionary	supports (general)	Does AIM33 carry reductase machinery?	Full FAD-binding FR-type + NAD-binding architecture, same as MCR1/CBR1	Sequence/ domain	Hi ox cla acc spe
Computed Needleman–Wunsch (this study)	structural/ evolutionary	qualifies	Is AIM33 a <i>close</i> ortholog of cyt-b5 reductases?	38.6% id vs CBR1, 32.4% vs MCR1 (CBR1–MCR1 = 33.6%) → paralog-level divergence	Sequence	Hi div we 1:1 tra
Motif scan (this study)	structural/ evolutionary	supports (general)	Are catalytic cofactor motifs intact?	CGPPKM (NAD) and GSGITP (FAD/NAD pyrophosphate) conserved	Sequence	Hi col bin acc ide
PANTHER PTHR19370:SF143	structural/ evolutionary	competing	Which subfamily / electron acceptor?	AIM33 in SF143 = "coenzyme Q6 reductase PGA3" , distinct from MCR1 (SF171)/CBR1 (SF184)	Phylogenetic	Me Co alt
PGA3 / UniProt Q12746 (subfamily seed)	review/ database + evolutionary	supports (family) / competing (location)	Does the characterized subfamily member have this activity?	PGA3 = "NADH-dependent cytochrome b5 reductase that reduces	<i>S. cerevisiae</i>	Me val MF ER

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Co lim
				coenzyme Q6 at the plasma membrane"; canonical cyt-b5 reductase reaction		
UniProt topology features	localization/structural	competing	Topology like classic cyt-b5 reductase?	3 TM helices (aa 15–35, 42–62, 180–200) — polytopic, vs single N-anchor of Mcr1/Cbr1	Sequence/topology	Me aty cla
P 16847258 (Kim et al. 2006)	localization	qualifies	Is AIM33 an integral membrane protein?	Experimentally-constrained polytopic membrane topology	<i>S. cerevisiae</i> , large-scale	Me thr
P 19300474 (Hess et al. 2009)	mutant phenotype	qualifies	Does AIM33 perform cyt-b5 reductase function?	Deletion increases mitochondrial genome loss / alters mito biogenesis	<i>S. cerevisiae</i> , deletion screen	Me do ph an
AlphaFold AF-Q04516-F1	structural/computational	supports (general)	Is the reductase fold intact/real?	Mean pLDDT = 87.25; confidently folded FNR-like module	Model	Hi ger fun
SGD GO:0003674 = ND	review/database	refutes (specific)	Is any experimental MF known?	SGD curated "No biological Data" for molecular function	<i>S. cerevisiae</i>	Hi ab exp ME

Provenance artifacts: `/tmp/aim33_evidence_matrix.csv` (computed table) plus the executed alignment/motif/AlphaFold/UniProt-query code and outputs in the iteration logs.

GO Curation Implications (leads — require curator verification)

GO ID	Aspect	Term	Basis	Recommended action (lead)
GO:0004128	MF	cytochrome-b5 reductase activity, acting on NAD(P)H	IBA from MCR1+PGA3; PGA3 experimentally has this activity; but AIM33 unconfirmed, SF143 hints CoQ, atypical topology	RETAIN as IBA but flag "inferred/unconfirmed." Acceptable as a phylogenetic MF. If a more conservative statement is preferred, generalize to the parent MF (below).
GO:0016491 / GO:0016651	MF	oxidoreductase / acting on NADH or NADPH	Domain + fold + conserved catalytic motifs	SUPPORTED — the safest evidence-backed MF generalization if a curator wants to avoid over-specificity.
GO:0005886	CC	plasma membrane	IBA carry-over from PGA3; conflicts with AIM33 mito phenotype; experimental location is only generic "Membrane"	REVISE / REMOVE. Highest-priority action. Replace with membrane (GO:0016020) or investigate mitochondrial/ER membrane.
GO:0006696	BP	ergosterol biosynthetic process	IBA carry-over from PGA3; no AIM33-specific evidence	TREAT AS NON-CORE / weak. Not supported by AIM33's own data.
(candidate) mitochondrial genome maintenance BP	BP	e.g., mitochondrial genome maintenance	P 19300474 loss-of-function	Candidate phenotype-based lead (IMP-type), curator judgement; note it is a downstream requirement, not necessarily direct.

Guidance note: "protein binding" is **not** used here — the family/domain evidence supports the more informative NAD(P)H:FAD oxidoreductase statement.

Mechanistic Scope

- **Immediate molecular function being tested:** NAD(P)H-dependent electron transfer via a bound FAD to an electron acceptor (cytochrome b5 and/or coenzyme Q6). The domain architecture, conserved `CGPPKM`/`GSGITP` motifs, and confident FNR-like fold directly support that AIM33 *can* bind NAD(P)H + FAD and act as an electron-transfer oxidoreductase.
- **Not established directly:** the identity of the physiological electron acceptor (cyt b5 vs CoQ6), catalytic turnover, and the subcellular membrane in which it acts.
- **Downstream / loss-of-function only (must not be conflated with MF):** the increased mitochondrial genome-loss phenotype
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is a genetic requirement in mitochondrial biogenesis/inheritance, not a molecular activity.
- **Pure database carry-over (no AIM33 data):** plasma-membrane localization and ergosterol biosynthesis, both inherited from PGA3 via IBA.

Conflicts and Alternatives

1. **Subfamily / acceptor alternative (competing):** AIM33 belongs to PANTHER SF143, whose namesake PGA3 reduces **coenzyme Q6**, not (only) cytochrome b5. The physiological acceptor for AIM33 could be ubiquinone; GO:0004128 may be over-specific relative to actual chemistry.
2. **Localization conflict (database carry-over):** the IBA CC "plasma membrane" and BP "ergosterol biosynthesis" match PGA3, but AIM33's experimental signal points to mitochondria (genome-loss phenotype) and only a generic "Membrane" location. This is classic single-anchor IBA package transfer.

3. **Topology divergence:** AIM33 is predicted polytopic (3 TM), unlike the mono-anchored soluble-domain architecture of the canonical NADH-cyt-b5 reductases Mcr1/Cbr1 — hinting at a functionally/locationally distinct role.
4. **Paralog landscape:** *S. cerevisiae* has ≥ 4 members (MCR1, CBR1, PGA3, AIM33). Functional/localization partitioning among paralogs means AIM33 need not replicate PGA3's plasma-membrane sterol role even if it shares the catalytic chemistry.
5. **Not a refutation:** because PGA3 (same subfamily, the IBA seed) genuinely has cyt-b5 reductase activity, the MF inference is *not* a spurious paralog over-annotation onto an unrelated protein — it is a reasonable but unconfirmed transfer.

Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
No direct enzymatic assay for AIM33	UniProt/QuickGO/SGD — SGD MF = ND	Determines whether GO:0004128 is direct vs inferred	In-vitro NAD(P)H-cytochrome b5 / CoQ6 reductase assay on purified AIM33
Electron-acceptor identity (cyt b5 vs CoQ6)	PANTHER subfamily, PGA3 function	Decides if GO:0004128 is precise or should be a CoQ-reductase term	Steady-state kinetics vs cyt b5 and vs ubiquinone analogs
True subcellular location	UniProt (generic "Membrane"), topology map, mito phenotype	Plasma-membrane CC likely wrong; affects CC curation	GFP/split-GFP or fractionation; check mitochondrial/ER membrane
NADH vs NADPH preference	Motifs present but not discriminating	Affects exact term (cyt-b5 reductase "acting on NAD(P)H")	Cofactor-preference kinetics
Mechanism linking to mito genome loss	P 19300474 (phenotype only)	Whether MF and phenotype are causally linked	Epistasis with Mcr1/Cbr1/Pga3; rescue by catalytic-dead AIM33

Discriminating Tests

1. **Purified-enzyme kinetics** with (a) cytochrome b5 and (b) coenzyme Q6/ubiquinone analogs, using NADH and NADPH — the single most decisive test of both activity and acceptor specificity.
2. **Catalytic-dead complementation**: test whether a FAD/NAD-binding-site mutant AIM33 (e.g., disrupting the `CGPPKM` or `GSGITP` motif) fails to rescue the mitochondrial genome-loss phenotype — links MF to phenotype.
3. **Localization by fractionation + high-resolution imaging** to test plasma-membrane vs mitochondrial/ER assignment.
4. **Comparative genetic interactions** (SGA) of *aim33Δ* vs *pga3Δ*, *mcr1Δ*, *cbr1Δ* to define functional overlap/partitioning within the paralog set.
5. **Structural superposition** of the AlphaFold AIM33 model onto solved Mcr1/human CYB5R3 to inspect acceptor-binding surface conservation.

Curation Leads (require curator verification)

- **Action on GO:0004128 (MF)**: *Retain as IBA but annotate as inferred/unconfirmed for AIM33.* Acceptable phylogenetic call; consider adding/keeping the generalized parent MF (oxidoreductase acting on NAD(P)H) as the conservative evidence-backed statement. Do **not** upgrade to an experimental evidence code.
- **Action on GO:0005886 (CC, plasma membrane)**: *Revise/remove* — likely mis-transferred from PGA3; replace with `membrane` (GO:0016020) pending localization data. **Highest-priority correction.**
- **Action on GO:0006696 (BP, ergosterol biosynthesis)**: *Treat as non-core / weak* — no AIM33-specific support.
- **Candidate reference to verify (disruption phenotype / mito role)**:

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— snippet to verify: "we have identified 100 proteins whose deficiency alters mitochondrial biogenesis and inheritance in *Saccharomyces cerevisiae*" (supports a mitochondrial genome-maintenance requirement, not a molecular-function assay).

- P 16847258**

— snippet to verify: *"present experimentally constrained topology models for 546 proteins"* (basis for the polytopic membrane topology).
- **Comparator to verify (family MF is genuine):** UniProt **Q12746 (PGA3)** — *"NADH-dependent cytochrome b5 reductase that reduces coenzyme Q6 at the plasma membrane"* — confirms the subfamily truly bears this activity, justifying the IBA while flagging the CoQ alternative.
- **Suggested curator questions:** (1) Is there any post-2009 experimental characterization of AIM33 enzymology or localization? (2) Should the MF be kept as-is or generalized given SF143's CoQ association? (3) Is the plasma-membrane CC defensible for AIM33 or a PGA3 carry-over?
- **Suggested experiments:** the discriminating tests above (enzyme kinetics vs cyt b5 and CoQ6; catalytic-dead rescue of the mito phenotype; localization).

Limitations

- The provided PubMed interface returned no yeast-genetics primary literature for AIM33/PGA3/AIM-screen queries; analysis relied on UniProt, QuickGO, InterPro, PANTHER, AlphaFold, and SGD primary records plus the two indexed primary papers

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- Sequence identity was computed with a simple Needleman–Wunsch (identity scoring), adequate for order-of-magnitude divergence but not a substitute for a curated multiple alignment / phylogeny.
- No wet-lab data could be generated; all activity/acceptor conclusions are inference-level and flagged as such.